

EDUCATION

- **BSDS**, Bachelor of Science in Data Science
Indian Institute of Technology (IIT), Madras 2024 - 2027 (expected to graduate)
- **MBBS**, Bachelor of Medicine, Bachelor of Surgery
(completed pre-clinical & research training; transitioned to AI research)
All India Institute of Medical Sciences (AIIMS), Rishikesh 2017 - On break

AI / ML RESEARCH OUTPUTS

- **Multimodal Learning & Vision-Language Models**
 - Garg P, Tan C. MetaLex: Probing Literal-to-Figurative Grounding in Vision-Language Models. CVPR 2026 Workshop on Any-to-Any Multimodal Learning (A2A-MML).
- **Mechanistic Interpretability & Robustness in Language Models**
 - Garg P. REPAIR: Diagnosing Spontaneous Context Restoration in Language Models. NeurIPS 2026 (Under Review). <https://openreview.net/pdf?id=Hnk3AHcqzG>
 - Garg P. A Control-Theoretic Account of Cognitive Effort in Language Models. NeurIPS 2025 Workshop (CogInterp). <https://neurips.cc/virtual/2025/loc/san-diego/129712>
 - Garg P. CoT Control: The Word Disappears but the Thought Does Not. LessWrong, 2026. [www.lesswrong.com]
- **Dynamical Systems & State-Space Modeling**
 - Garg P, Nema V. Modeling neural dynamics using periodic structured state space models. NeurIPS 2024 WiML Workshop. <https://neurips.cc/virtual/2024/109052>
- **Representation Learning in Neuroscience Data**
 - de Oliveira E.F., Garg P, et al. Generalized Contrastive PCA. *PLOS Computational Biology*, 2025. DOI: 10.1371/journal.pcbi.1012747

AI/ML RESEARCH EXPERIENCE

- **Research Intern**, A*STAR Centre for Frontier AI Research January 2026 - March 2026
Advisor: Dr Cheston Tan, PhD
Project: Literal-to-Figurative Grounding in Vision-Language Models (MetaLex)
 - Developed *MetaLex*, a benchmark for evaluating literal-to-figurative grounding in VLMs under controlled, semantically matched inputs.
 - Demonstrated systematic modality asymmetry: identical concepts are less accessible from visual inputs than from linguistic inputs, despite equivalent content.
 - Designed controlled protocols (weak-captioner, joint conditioning) to disentangle representational access from informational effects in cross-modal reasoning.
 - Identified structured, directional modality gaps and non-additive cross-modal interactions, including linguistic scaffolding and visual interference.
- **Independent research** July 2025- December 2025
Project: A Control-Theoretic Account of Cognitive Effort in Language Models
 - Modeled transformer residual streams as time-varying linear dynamical systems using control theory.
 - Introduced energy-based metrics (geometric difficulty, efficiency) via controllability Gramians.
 - Showed post-training compresses controllable subspaces, increasing control cost across tasks.
 - Found open-ended conversational questions are geometrically harder than open-ended math questions, contradicting human intuition.
 - Accepted at the NeurIPS 2025 First Workshop on CogInterp: Interpreting Cognition in Deep Learning Model as a single-author manuscript; Extended version in preparation for submission to ICML.
- **Independent research** January 2025- October 2025
Project: REPAIR: Diagnosing Spontaneous Context Restoration in Language Models
 - Studied spontaneous context restoration under controlled, label-preserving symbolic corruptions in small transformers.
 - Introduced layer-wise diagnostics (Repair Difference, Token Agreement) to localize internal self-repair dynamics.
 - Identified a consistent mid-to-late-layer repair “elbow” separating error propagation from correction.
 - Linked repairability to local linearity via scale tests and causal residual-gain interventions.
 - Replicated repair and curvature patterns across multiple pretrained LLMs on NLP corruption tasks.
 - Showed that moderate corruption fine-tuning improves robustness, while heavy corruption degrades repair.
 - Sole-author manuscript under consideration at ICML.

- **Independent research**

August 2024- October 2024

Project: Modeling neural dynamics using periodic structured state space mode

- Introduced a periodic structured state-space model (pS4) to capture low-dimensional neural dynamics.
- Modeled hippocampal population activity as a linear time-varying control system aligned to neural oscillations.
- Demonstrated stable training, noise reduction, and accurate next-step prediction of latent dynamics.
- Interpreted learned dynamics via control cost and feedback gains before and after stimulus perturbations.
- Lead-authored poster accepted at the NeurIPS 2024 Women in Machine Learning (WiML) Affinity Event.

- **Research Intern (remote)**, Albert Einstein College Of Medicine

Aug 2022 - June 2024

Advisor: Dr. Luke Sjulson, MD, PhD

Project: Developing novel linear dimensionality reduction algorithm, generalized contrastive principal component analysis (gcPCA)

- Developed a linear dimensionality reduction method to isolate condition-specific representational subspaces in high-dimensional data.
- Analyzed large-scale electrophysiology data from the Allen Institute Brain Observatory and benchmarked gcPCA against PCA using supervised classification.
- Demonstrated generality across modalities, including MNIST, facial image datasets, and electrophysiological recordings.
- Analyzed single-cell RNA-seq data from the Allen Institute, including differential gene expression.
- Designed biologically grounded behavioral tasks and trained continuous-time recurrent neural networks (CTRNNs) with ablation and representational analyses.
- Co-author on a peer-reviewed publication in *PLOS Computational Biology*; work accepted at SfN 2024.

NEUROSCIENCE RESEARCH EXPERIENCE

- **Research Intern**, Max Planck Institute for Biological Intelligence

October 2024 – October 2025

Advisor: Dr. Lisa Fenk, PhD

Project: Investigating the role of scene statistics in differential retinal saccadic responses.

- Developed automated algorithms to detect retinal saccades and drift from fly eye-tracking and ball-walking data.
- Quantified natural scene statistics using a quantum-catch-weighted local binary pattern entropy measure.
- Conducted tethered fly behavioural experiments with controlled and naturalistic visual stimuli.
- Analysed how spontaneous retinal movements adapt to scene statistics and reshape spatiotemporal visual signals

- **Research Intern (remote)**, Southern Illinois Institute

June 2024 - October 2024

Advisor: Dr. Duda Kvitsiani, PhD and Dr. Ebru Demir, PhD

Project: Social interaction-water reward trade-off in the orbitofrontal cortex

- Conducted data analysis of calcium imaging time-series data using Python
- Constructed and trained of recurrent neural networks to discover underlying attractor dynamics in the data
- Trained support vector machines to deduce encoding and memory axes (vectors orthogonal to the hyperplane)

- **Independent research**

June 2023–December 2023

Project: Signatures of criticality in the hippocampus during awake and sleep epochs in rodents

- Formulated a hypothesis linking hippocampal sharp-wave ripples to neuronal criticality using a public LFP dataset.
- Detected neuronal avalanches and sharp-wave ripples from LFP signals using scale-invariant thresholding.
- Quantified deviation from criticality across awake and sleep states using avalanche and branching metrics.
- Scaled analyses via distributed execution on Microsoft Azure HPC using Python.
- Authored a preprint and presented findings at the SfN 2024 meeting.

- **Independent research**

June 2022–August 2023 (Remote)

Project: Modelling Cortical Up-Down State Switching by Astrocytes

- Built a recurrent neural network (RNN) rate model of cortical Up-Down state dynamics with excitatory, inhibitory, and astrocyte populations.
- Extended neuron-astrocyte RNN models to include hyperpolarizing gliotransmission.
- Performed linear stability analysis to identify bistable regimes supporting spontaneous state transitions.
- Applied global sensitivity analysis (Morris method) to determine parameters governing RNN dynamics.
- Found excitatory thresholds and adaptation dominate behavior, with lower sensitivity to coupling strengths.
- Corresponding author on a preprint; work presented at the Fourth International Symposium on the Mathematics of Neuroscience.

- **Research Fellow**, California Institute of Technology, Pasadena, USA

Aug 2021 - Oct 2021

Advisor: Dr. Paul Sternberg, PhD

Project: Dye filling in the head neurons of *Steinernema hermaphroditum* and the neuroanatomy characterization

- Developed and optimized a Dil fluorescence staining protocol to label head sensory neurons in *Steinernema hermaphrodi-*

tum.

- Quantitatively evaluated staining conditions (dye concentration, buffer, incubation time) and compared performance across protocols.
- Identified and anatomically characterized amphid and inner labial neurons, establishing positional homology with *C. elegans*.
- Demonstrated dye-filling resistance in infective juveniles, consistent with species-specific sensory adaptations.
- Authored the manuscript as first author, resulting in a publication, and presented the work at an international symposium.

MEDICAL & HUMANITIES RESEARCH EXPERIENCE

- **Independent research** 2023
Project: Exploring the Relationship Between Popular Culture and Perceptions of Social Standing in STEM Careers
 - Designed an anonymized, multinational survey study on popular culture and STEM career perceptions.
 - Operationalized social and professional associations with STEM media representations across India and the U.S.
 - Estimated direct and indirect effects using structural equation modeling (SEM) in R (lavaan).
 - Showed that popular culture consumption shapes perceived social standing of STEM careers.
 - Authored a preprint and presented as first author at the Cognitive Science Society Meeting.
- **Research Intern**, University of Minnesota (remote) March 2022 – June 2022
Advisor: Dr. Olihe Okoro, PhD
Project: Awareness and Acceptability of the Malaria Vaccine in India
 - Collected and analysed nationwide survey data on malaria vaccine awareness in India.
 - Contributed to manuscript preparation; co-author on a peer-reviewed publication.
- **Research Assistant**, All India Institute of Medical Sciences (AIIMS) 2021
Advisor: Dr. Rajesh Kathrotia, MD
Project: Sleep Hygiene, Excessive Daytime Sleepiness, and Sleep Quality Across Socioeconomic Strata
 - Designed and analyzed a stratified population study on sleep and socioeconomic status.
 - First author on a peer-reviewed publication; presented findings internationally.

EARLY RECOGNITIONS

- COSYNE Undergraduate Travel Award 2024, 2023
- Openscope Program (Allen Institute), Stage 2 Selection 2023
- NIH SPARC FAIR Codeathon – 3rd Prize (\$7,000) 2023
- Competitive grants secured under my leadership for Project Encephalon: IBRO APRC Grant (€1,250), IndiaBioscience Outreach Grant (\$1,300), and Falling Walls Engage Finalist 2020–2022
- Summer Undergraduate Research Fellowship, Caltech 2021
- Additional early honours include KVPY (DST, India) and IISF Travel Grant.

OTHER SELECTED PUBLICATIONS & PRESENTATIONS

- **Garg P.** Mixed signatures for subcritical dynamics in rodent hippocampus during sleep and awake epochs. Society for Neuroscience (SfN) Meeting, 2024. DOI: 10.1101/2023.10.30.564597
- Verma J, **Garg P.** Computational Modeling of Hyperpolarising Astrocytic Influence on Cortical Up-Down State Transitions. Fourth International Symposium on the Mathematics of Neuroscience, 2023. DOI: 10.1101/2023.10.16.562461
- **Garg P.** Exploring the relationship between popular culture and perceptions of social standing in STEM careers. Cognitive Science Society Meeting, 2023. <https://escholarship.org/uc/item/05m8h9t1>; DOI: 10.31234/osf.io/c4p3y
- **Garg P**, Tan C.H., Sternberg P.W*. DiI staining of sensory neurons in the entomopathogenic nematode *Steinernema hermaphroditum*. *microPublication Biology*, 2022. DOI: 10.17912/micropub.biology.000516
- Singhal C, Aremu T.O., **Garg P**, et al. Awareness of the Malaria Vaccine in India. *Cureus*, 2022. DOI:10.7759/cureus.29210
- Garg, P. and Gupta, S. (2022) "IoT-Based Disease Prediction," in Smart and Secure Internet of Healthcare Things. 1st edn. Florida: CRC Press. DOI: 10.1201/9781003239895-2

SELECTED LEADERSHIP EXPERIENCE

- **Founder and Managing Trustee**, Project Encephalon Foundation June 2020 – Present

Project Encephalon is an international, trainee-led non-profit organization for neuroscience enthusiasts, facilitating seminars, symposia, conferences, mentorship sessions, and more.

- Organized global neuroscience outreach programs and events for resource-limited settings.
- Built a diverse team of international volunteers and managed operations spanning multiple time zones.
- Established collaborations with academic and professional organizations to expand program reach.
- **Student Editor**, International Journal of Medical Students *August 2022 – February 2023*
 - Reviewed and edited scientific manuscripts for publication in a globally recognized journal.
 - Mentored aspiring student researchers in improving their writing and presentation skills.
 - Streamlined the peer-review process to enhance efficiency and uphold publication standards.